

Table S2. Full near-eye metric dictionary and distributions

Metric descriptive summary (near eye)													
Metric ¹	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Duration													
Time above 1,000 lx meEDI Bright-light exposure duration; relevant to daytime alerting and circadian entrainment.	HH:MM	00:41 (00:11, 01:33) <i>01:10 ± 01:28</i> N=141; d=816	01:18 (00:31, 03:24) <i>02:00 ± 01:54</i> N=13; d=78	01:18 (00:29, 02:12) <i>01:33 ± 01:28</i> N=13; d=78	01:03 (00:20, 02:32) <i>01:48 ± 02:14</i> N=18; d=107	00:40 (00:07, 01:25) <i>01:06 ± 01:22</i> N=26; d=150	00:39 (00:10, 01:46) <i>01:09 ± 01:18</i> N=10; d=60	00:27 (00:06, 00:53) <i>00:38 ± 00:47</i> N=23; d=129	00:30 (00:06, 01:04) <i>00:44 ± 00:51</i> N=17; d=101	00:19 (00:09, 00:43) <i>00:32 ± 00:33</i> N=6; d=32	00:43 (00:17, 01:14) <i>00:58 ± 00:57</i> N=15; d=81	Symlog (base 10; threshold 1)	
Time above 250 lx meEDI during wake Waking time in recommended daytime light; relevant to alertness, entrainment, and subsequent sleep.	HH:MM	02:25 (00:51, 04:37) <i>03:00 ± 02:29</i> N=141; d=737	04:06 (02:30, 06:15) <i>04:21 ± 02:24</i> N=13; d=73	03:29 (02:03, 05:16) <i>03:39 ± 02:25</i> N=13; d=64	03:42 (02:05, 05:58) <i>04:05 ± 02:51</i> N=18; d=90	02:01 (00:47, 04:20) <i>02:52 ± 02:39</i> N=26; d=141	02:38 (00:55, 04:19) <i>02:57 ± 02:19</i> N=10; d=54	02:43 (00:54, 04:43) <i>03:02 ± 02:19</i> N=23; d=113	01:30 (00:41, 03:26) <i>02:09 ± 01:57</i> N=17; d=94	01:39 (00:33, 02:40) <i>01:48 ± 01:21</i> N=6; d=30	01:14 (00:25, 02:37) <i>01:45 ± 01:46</i> N=15; d=78	Symlog (base 10; threshold 1)	
Time below 10 lx meEDI before sleep Low-light time before bed; limits evening melatonin suppression and circadian delay.	HH:MM	01:53 (01:01, 02:35) <i>01:51 ± 01:02</i> N=139; d=655	02:16 (01:30, 02:39) <i>02:05 ± 00:53</i> N=13; d=67	01:50 (01:21, 02:52) <i>01:59 ± 00:57</i> N=12; d=50	01:42 (00:53, 02:17) <i>01:41 ± 00:58</i> N=18; d=84	01:52 (01:04, 02:25) <i>01:52 ± 01:06</i> N=26; d=132	01:43 (00:42, 02:32) <i>01:45 ± 01:10</i> N=10; d=52	01:43 (00:58, 02:50) <i>01:50 ± 01:05</i> N=23; d=105	01:31 (00:49, 02:15) <i>01:38 ± 01:00</i> N=17; d=82	02:16 (01:15, 02:39) <i>01:56 ± 01:00</i> N=6; d=25	02:30 (01:28, 02:52) <i>02:09 ± 00:57</i> N=14; d=58	Symlog (base 10; threshold 1)	
Time below 1 lx meEDI during sleep Darkness during sleep; supports nocturnal melatonin and an undisturbed sleep environment.	HH:MM	07:08 (05:58, 08:15) <i>07:04 ± 01:57</i> N=141; d=778	07:08 (06:03, 08:18) <i>07:12 ± 01:46</i> N=13; d=76	07:14 (05:56, 08:06) <i>07:01 ± 01:50</i> N=13; d=68	07:08 (05:56, 08:16) <i>07:05 ± 01:50</i> N=18; d=95	06:54 (05:51, 07:46) <i>06:47 ± 01:30</i> N=26; d=150	05:52 (04:41, 07:03) <i>05:58 ± 01:58</i> N=10; d=60	08:06 (07:10, 09:10) <i>08:02 ± 01:57</i> N=23; d=118	06:45 (05:08, 08:15) <i>06:40 ± 02:07</i> N=17; d=99	06:56 (06:06, 07:36) <i>06:39 ± 01:43</i> N=6; d=31	07:31 (06:31, 09:00) <i>07:35 ± 02:15</i> N=15; d=81	Symlog (base 10; threshold 1)	
Longest period above 250 lx meEDI Longest sustained bright-light bout; captures continuity of daytime circadian stimulation.	HH:MM	00:38 (00:17, 01:12) <i>00:55 ± 01:00</i> N=141; d=816	00:57 (00:28, 01:27) <i>01:12 ± 01:02</i> N=13; d=78	00:48 (00:23, 01:20) <i>01:09 ± 01:15</i> N=13; d=78	01:00 (00:21, 01:40) <i>01:20 ± 01:31</i> N=18; d=107	00:35 (00:13, 01:06) <i>00:50 ± 00:54</i> N=26; d=150	00:38 (00:17, 01:32) <i>01:01 ± 01:02</i> N=10; d=60	00:38 (00:16, 01:04) <i>00:45 ± 00:36</i> N=23; d=129	00:34 (00:14, 00:55) <i>00:43 ± 00:41</i> N=17; d=101	00:28 (00:16, 00:43) <i>00:30 ± 00:20</i> N=6; d=32	00:23 (00:15, 00:49) <i>00:38 ± 00:45</i> N=15; d=81	Symlog (base 10; threshold 1)	
Dynamics													
Interdaily stability Day-to-day regularity of the light–dark pattern; higher regularity supports circadian stability.	—	0.308 (0.248, 0.38) <i>0.318 ± 0.095</i> N=141; d=816	0.311 (0.272, 0.367) <i>0.315 ± 0.078</i> N=13; d=78	0.258 (0.239, 0.389) <i>0.288 ± 0.094</i> N=13; d=78	0.264 (0.219, 0.319) <i>0.274 ± 0.074</i> N=18; d=107	0.329 (0.263, 0.424) <i>0.344 ± 0.107</i> N=26; d=150	0.244 (0.218, 0.277) <i>0.252 ± 0.058</i> N=10; d=60	0.352 (0.321, 0.455) <i>0.368 ± 0.092</i> N=23; d=129	0.278 (0.254, 0.394) <i>0.323 ± 0.104</i> N=17; d=101	0.342 (0.315, 0.473) <i>0.37 ± 0.118</i> N=6; d=32	0.293 (0.255, 0.354) <i>0.299 ± 0.062</i> N=15; d=81	Identity	
Intradaily variability Within-day fragmentation of light exposure; higher values indicate less consolidated light–dark input.	—	1.253 (0.93, 1.502) <i>1.229 ± 0.389</i> N=141; d=816	0.997 (0.818, 1.388) <i>1.088 ± 0.343</i> N=13; d=78	1.193 (0.96, 1.389) <i>1.238 ± 0.345</i> N=13; d=78	1.182 (0.928, 1.49) <i>1.177 ± 0.407</i> N=18; d=107	1.154 (0.801, 1.442) <i>1.125 ± 0.354</i> N=26; d=150	1.48 (0.869, 1.642) <i>1.332 ± 0.477</i> N=10; d=60	1.251 (0.999, 1.497) <i>1.251 ± 0.413</i> N=23; d=129	1.359 (0.95, 1.546) <i>1.307 ± 0.415</i> N=17; d=101	1.399 (1.282, 1.483) <i>1.355 ± 0.256</i> N=6; d=32	1.324 (1.029, 1.595) <i>1.344 ± 0.413</i> N=15; d=81	Identity	
Exposure history													
meEDI dose Intensity–duration-weighted melanopic exposure; summarizes cumulative non-visual retinal light input.	klx·h	4.96 (1.936, 12.313) <i>10.358 ± 16.216</i> N=141; d=761	9.296 (3.892, 21.895) <i>19.923 ± 27.784</i> N=13; d=71	10.495 (3.855, 17.395) <i>15.881 ± 20.395</i> N=13; d=74	6.751 (2.469, 14.435) <i>13.809 ± 23.711</i> N=18; d=97	3.862 (1.411, 10.217) <i>7.719 ± 9.847</i> N=26; d=137	7.11 (1.897, 15.789) <i>10.055 ± 11.237</i> N=10; d=55	3.317 (1.703, 6.29) <i>6.115 ± 8.79</i> N=23; d=125	4.35 (1.791, 10.625) <i>7.49 ± 8.898</i> N=17; d=96	2.195 (1.339, 4.117) <i>3.281 ± 2.573</i> N=6; d=30	6.045 (2.49, 14.907) <i>10.016 ± 10.799</i> N=15; d=76	Symlog (base 10; threshold 1)	
Level													
Mean meEDI Geometric average of daily meEDI values, including zeros; summarizes overall exposure while reducing peak influence.	lx	5.154 (2.831, 9.225) <i>7.587 ± 9.496</i> N=141; d=816	7.569 (4.55, 11.613) <i>9.171 ± 7.111</i> N=13; d=78	6.659 (4.18, 11.674) <i>8.274 ± 6.558</i> N=13; d=78	7.704 (3.081, 13.924) <i>12.017 ± 18.865</i> N=18; d=107	5.232 (3.224, 8.188) <i>7.221 ± 6.33</i> N=26; d=150	9.013 (4.011, 15.089) <i>12.102 ± 12.013</i> N=10; d=60	3.505 (1.469, 6.211) <i>4.166 ± 3.272</i> N=23; d=129	5.857 (3.726, 7.986) <i>7.778 ± 6.946</i> N=17; d=101	4.325 (3.585, 6.77) <i>5.248 ± 2.85</i> N=6; d=32	2.144 (0.744, 4.042) <i>3.019 ± 2.896</i> N=15; d=81	Symlog (base 10; threshold 1)	

Metric descriptive summary (near eye)													
Metric ¹	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Brightest 10 h mean Mean of the brightest 10 hours; reflects the strength of the main daytime light episode.	lx	110.566 (41.513, 243.212) 214.866 ± 508.56 N=141; d=816	252.689 (104.514, 461.953) 399.302 ± 518.102 N=13; d=78	175.974 (78.625, 299.106) 262.206 ± 329.201 N=13; d=78	156.184 (67.783, 339.731) 422.54 ± 1,209.616 N=18; d=107	100.057 (33.255, 209.904) 179.372 ± 241.53 N=26; d=150	130.78 (54.044, 259.726) 201.947 ± 194.042 N=10; d=60	101.102 (24.355, 218.457) 136.312 ± 134.07 N=23; d=129	83.24 (53.17, 151.786) 120.172 ± 101.545 N=17; d=101	68.143 (45.331, 132.458) 96.735 ± 72.311 N=6; d=32	46.66 (9.242, 103.175) 82.489 ± 115.07 N=15; d=81	Symlog (base 10; threshold 1)	
Darkest 10 h mean Mean of the darkest 10 hours; lower values during the biological night favour melatonin preservation and sleep.	lx	0.103 (0.02, 0.253) 0.243 ± 0.513 N=141; d=816	0.08 (0.016, 0.191) 0.183 ± 0.299 N=13; d=78	0.1 (0.035, 0.23) 0.2 ± 0.332 N=13; d=78	0.14 (0.036, 0.282) 0.223 ± 0.299 N=18; d=107	0.149 (0.074, 0.309) 0.273 ± 0.465 N=26; d=150	0.325 (0.119, 0.71) 0.515 ± 0.653 N=10; d=60	0.022 (0, 0.08) 0.054 ± 0.075 N=23; d=129	0.152 (0.052, 0.481) 0.475 ± 0.882 N=17; d=101	0.133 (0.055, 0.309) 0.411 ± 1.044 N=6; d=32	0.01 (0, 0.092) 0.059 ± 0.104 N=15; d=81	Symlog (base 10; threshold 1)	
Spectrum													
Melanopic daylight efficacy ratio Mean of viable one-minute melEDI/illuminance ratios; indicates melanopic efficacy relative to visual light.	—	0.724 (0.643, 0.795) 0.724 ± 0.116 N=137; d=687	0.788 (0.725, 0.872) 0.796 ± 0.162 N=13; d=74	0.743 (0.699, 0.796) 0.743 ± 0.077 N=13; d=65	0.781 (0.675, 0.853) 0.768 ± 0.118 N=18; d=97	0.65 (0.592, 0.728) 0.667 ± 0.098 N=26; d=144	0.724 (0.652, 0.773) 0.72 ± 0.093 N=10; d=55	0.668 (0.629, 0.721) 0.673 ± 0.07 N=22; d=83	0.747 (0.634, 0.834) 0.735 ± 0.122 N=17; d=91	0.719 (0.661, 0.761) 0.706 ± 0.067 N=6; d=30	0.755 (0.729, 0.82) 0.758 ± 0.097 N=12; d=48	Identity	
Timing													
Midpoint of the brightest 10 hours Centre time of the brightest 10 hours; indexes the main daily circadian light cue.	HH:MM clock time	13:44 (12:48, 15:00) 13:56 ± 01:51 N=141; d=816	13:16 (12:28, 14:10) 13:20 ± 01:33 N=13; d=78	14:12 (13:23, 15:18) 14:22 ± 01:51 N=13; d=78	13:52 (12:54, 15:04) 14:00 ± 01:42 N=18; d=107	13:38 (12:47, 14:44) 13:52 ± 01:42 N=26; d=150	14:54 (12:50, 16:03) 14:31 ± 02:14 N=10; d=60	13:44 (13:06, 15:20) 14:10 ± 01:51 N=23; d=129	14:10 (13:32, 15:21) 14:25 ± 01:37 N=17; d=101	12:34 (11:49, 14:02) 12:55 ± 01:44 N=6; d=32	12:49 (12:06, 13:39) 13:03 ± 01:58 N=15; d=81	Circular clock	
Midpoint of the darkest 10 hours Centre time of the darkest 10 hours; indexes the main daily darkness cue.	HH:MM clock time	02:54 (02:01, 03:50) 02:58 ± 01:41 N=141; d=816	02:02 (01:25, 02:48) 02:03 ± 01:07 N=13; d=78	02:55 (02:01, 03:27) 02:50 ± 02:05 N=13; d=78	02:32 (01:45, 03:22) 02:37 ± 01:19 N=18; d=107	03:08 (02:11, 04:10) 03:16 ± 01:34 N=26; d=150	02:37 (01:54, 04:17) 03:10 ± 02:00 N=10; d=60	03:39 (02:43, 04:34) 03:34 ± 01:49 N=23; d=129	03:31 (02:33, 04:14) 03:26 ± 01:25 N=17; d=101	01:45 (01:05, 03:01) 02:10 ± 01:27 N=6; d=32	02:33 (02:04, 03:17) 02:36 ± 01:36 N=15; d=81	Circular clock	
First light timing above 250 lx melEDI First waking bright-light exposure; morning timing can advance circadian phase and promote alertness.	HH:MM clock time	09:08 (08:02, 10:40) 09:26 ± 02:22 N=140; d=727	07:57 (06:56, 08:33) 08:03 ± 01:32 N=13; d=74	09:04 (08:18, 09:52) 09:17 ± 01:56 N=13; d=68	08:36 (07:10, 10:13) 08:45 ± 02:21 N=18; d=93	09:43 (08:13, 10:53) 09:45 ± 02:21 N=25; d=133	09:36 (07:58, 10:53) 09:39 ± 03:12 N=10; d=56	09:32 (08:40, 11:21) 10:05 ± 02:18 N=23; d=118	10:01 (09:06, 11:52) 10:26 ± 02:28 N=17; d=95	08:11 (07:26, 09:57) 08:36 ± 01:30 N=6; d=29	08:43 (08:01, 10:10) 09:19 ± 01:52 N=15; d=61	Circular clock	
Last light timing above 250 lx melEDI Last bright-light exposure; later timing may delay circadian phase and sleep onset.	HH:MM clock time	18:08 (16:27, 19:42) 18:03 ± 02:36 N=141; d=687	18:41 (17:38, 19:38) 18:39 ± 01:38 N=13; d=73	18:30 (17:37, 19:59) 18:37 ± 02:08 N=13; d=62	19:34 (18:04, 20:42) 19:16 ± 02:28 N=18; d=89	17:37 (15:35, 19:09) 17:18 ± 02:44 N=26; d=121	19:50 (18:37, 20:50) 19:33 ± 02:11 N=10; d=53	17:48 (16:46, 19:19) 18:01 ± 02:21 N=23; d=106	18:45 (17:02, 19:38) 18:23 ± 02:15 N=17; d=89	15:45 (13:20, 16:38) 15:22 ± 02:38 N=6; d=28	15:54 (14:34, 17:25) 15:46 ± 02:13 N=15; d=66	Circular clock	
Mean timing of exposure above 250 lx melEDI Average bright-light timing; summarizes the phase of daily circadian stimulation.	HH:MM clock time	13:29 (12:29, 14:39) 13:31 ± 01:52 N=141; d=742	13:20 (12:29, 14:24) 13:21 ± 01:25 N=13; d=72	13:52 (13:13, 15:06) 14:02 ± 01:41 N=13; d=71	13:44 (12:28, 14:55) 13:51 ± 01:56 N=18; d=93	13:17 (12:17, 14:31) 13:24 ± 01:45 N=26; d=138	14:04 (12:58, 15:25) 14:09 ± 02:22 N=10; d=55	13:43 (12:55, 14:29) 13:47 ± 01:33 N=23; d=117	13:58 (12:54, 14:58) 14:04 ± 01:39 N=17; d=94	11:30 (10:28, 12:38) 11:26 ± 01:36 N=6; d=31	12:17 (11:04, 13:04) 12:11 ± 01:36 N=15; d=71	Circular clock	

Median (25th percentile, 75th percentile), *circular or arithmetic mean ± standard deviation*, N=participants; d=participant-days with a finite value for that metric.

¹ The brief meaning and relevance notes describe established physiological constructs; this descriptive table does not estimate individual health effects.

² Scaling describes the distribution column only; printed values remain on their stated scale. Symlog uses base 10 with a linear region through 1 in the stated unit; Identity is linear; Circular clock unwraps values around the metric-specific clock centre.

³ Red lines indicate site medians. Symlog and circular-clock rows are transformed only for plotting.

Supplementary Table S2. Definitions, exact support, median (interquartile range) summaries, and density displays for the 17 near-eye personal light-exposure metrics overall and by site. Values use three decimal places where applicable. Metrics are distinct summaries and should not be interpreted as interchangeable outcomes.